Individual Diagnostic Requirement Specification

CN200S ECM (1.5L)

Version: V1.0

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Requirement wording conventions

Within this document the following conventions are applied when describing specific compliance requirements:

The word “Should” denotes a preference or desired conformance.

The word “Shall” is used in the following ways:

1. To state a binding requirement on the special network, or the ECUs which comprise the network, which is verifiable by external manipulation and/or observation of an ECU or the network.

2. To state a binding requirement upon an ECU’s requirements document that is verifiable through a review of the document.

The word “Must” is used to state a binding requirement upon ECUs on the network which have a corresponding ECU requirements document. These requirements will be verified as part of the component verification.

NOTE: In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence.

NOTE: Nothing in this specification supersedes applicable laws and regulations unless a specific exemption has been obtained.

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History record of the document

The following table contains the revision history for this specification.
Changes will not be listed if corrections of the spelling or formal changes are made.

Version#	Date	Editor	Description of the revision
V1.0	2014-7-21	Liao wenqing	Initial release

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1 Introduction

1.1 Scope

The detailed description about this chapter can be found in chapter 1.1 of the document *CN200S General Diagnostic Requirement Specification*.

NOTE: For CN200S vehicle, ECM need to realize the software download function, the detailed requirements about software download can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

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2 Normative references

This document references ISO,SGMW and other publications as listed in the following sub sections.

2.1 ISO Publications

ISO 11898-1:2003 Road Vehicles - Controller Area Networks (CAN) - Part 1: Data link layer and physical signalling

ISO 11898-2:2003 Road Vehicles - Controller Area Networks (CAN) – Part 2: High-speed medium access unit

ISO 15765-1:2004 Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information

ISO 15765-2:2004 Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services

ISO 15765-3:2004 Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of unified diagnostic services (UDS on CAN)

ISO 14229-1:2006 Road Vehicles - Unified diagnostic services (UDS) - Specification and requirements

ISO 15031-6:2005 Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 6: Diagnostic trouble code definitions

2.2 SGMW Standard

CN200S General Diagnostic Requirement Specification
BT/SGMWJ 0858:2013 高速 CAN 物理层规范

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3 Terms, definitions and abbreviations

3.1 ISO Terms

This document makes use of terms defined in the ISO 15765-2, ISO 15765-3 and ISO 14229-1.

3.2 Specific Terms Defined By This Document

Table 3-1 Specific Terms Defined By This Document

Abbreviation/缩写	Description/描述
msec	**mili**l**sec**ond
CANId	**CAN I**dentifier
ABS/ESP	**A**nti-lock **B**rake **S**ystem/**E**lectronic **S**tability **P**rogram
TPMS	**T**ire **P**ressure **M**onitor **S**ystem
SDM	**R**estraint **S**ensing and**D**iagnostic **M**odule
DCU	**D**omain **C**ontroller **U**nit
BCM	**B**ody **C**ontrol **M**odule
PEPS	**P**assive **E**ntry**&P**assive **S**tart
EPS	**E**lectronic **P**ower **S**teering
AC	**Auto fixed-temp AC**
IC	**I**nstrument **C**luster
ICE	**I**n **C**ar **E**ntertainment
ECM	**E**ngine **C**ontrol **M**odule
TCM	**T**ransmission **C**ontrol **M**odule

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4 General requirement

All the general requirements, with which the tester and ECM shall comply, can be found in section 4 of the document *CN200S General Diagnostic Requirement Specification*.

4.1 Diagnostic CAN identifiers definition

The following table provides an overview about the definition of ECM diagnostic CAN identifier.

Table 4-1 ECM diagnostic CAN identifier

ECU		Functionally addressed request from tester to ECUs	Physically addressed request from tester to ECU	Response from ECU to tester
Name	Emission-related OBDECU (YES/NO)			
ECM	YES	0x7DF	0x7E0	0x7E8

4.2 ECU Diagnostic Address definition

The following table provides an overview about the definition of ECM diagnostic Address.

Table 4-2 ECM Diagnostic Address

ECU Name	Description	Value	Comments
ECM	ECUDiagnosticAddress	0x11	ECM diagnostic address, which is defined in *CN200S General Diagnostic Requirement Specification*.

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5 Physical layer requirement

All the physical layer requirements, with which the tester and ECM shall comply, can be found in section 5 of the document *CN200S General Diagnostic Requirement Specification*.

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6 Data link layer requirement

All the data link layer requirements, with which the tester and ECM shall comply, can be found in section 6 of the document *CN200S General Diagnostic Requirement Specification*.

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7 Network layer requirement

All the network layer requirements, with which the tester and ECM shall comply, can be found in section 7 of the document *CN200S General Diagnostic Requirement Specification*.

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8 Application and session layers requirement

All the application and session layer requirements, with which the tester and ECM shall comply, can be found in section 8 of the document *CN200S General Diagnostic Requirement Specification*.

8.1 Diagnostic Service Table Overview

The table followed (Table 8-1) shows the services which are allowed during the defaultSession, programmingSession and extendedSession of ECM. ECM is allowed to support more other diagnostic services per session after confirmed by SGMW.

Table 8-1 Diagnostic Service Table Overview

Service Name 服务名称	SID (Hex)	Sub-function/ 子功能 (Hex)	Session/会话						Security Access[1]/ 安全访问		
			Default 默认		Programming 编程		Extended 扩展		LV1 等级 1	LV 2 等级 2	LV 3 等级 3
			Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址			
Diagnostic and communication management functional unit/诊断与通信管理功能单元											
DiagnosticSessionControl	10	01	√[2]	√	√	√	√	√	×[2]	×	×
		02	√	×	√	×	√	×	×	×	×
		03	√	√	√	√	√	√	×	×	×
ECUReset	11	01	√	√	√	√	√	√	×	×	×
		02	√	√	×	×	√	√	×	×	×
SecurityAccess	27	01/02	×	×	×	×	√	×	×	×	×
		03/04	×	×	×	×	√	×	×	×	×
		05/06	×	×	√	×	×	×	×	×	×
CommunicationControl	28	00/03	×	×	×	×	√	√	×	×	×
TesterPresent	3E	00/80	√	√	√	√	√	√	×	×	×
ControlDTCSetting	85	01/02	×	×	×	×	√	√	×	×	×
ProgrammingMode	A5	01/03	×	×	×	×	√	√	×	×	×
Data transmission functional unit/数据传输功能单元											
ReadDataByIdentifier	22	-	√	√[3]	√	×	√	√[3]	×	×	×
DynamicallyDefineDataIdentifier	2C	-	√	×	×	×	√	×	×	×	×
WriteDataByIdentifier	2E	-	×	×	√	×	√	×	√[4]	×	×
Stored data transmission functional unit/存储数据传输功能单元											
ClearDiagnosticInformation	14	-	√	√	×	×	√	√	×	×	×
ReadDTCInformation	19	01/02/04/06/0A	√	√	×	×	√	√	×	×	×
InputOutput control functional unit/输入输出控制功能单元											
InputOutputControlByIdentifier	2F	-	×	×	×	×	√	×	×	×	×
Remote activation of routine functional unit/程序远程激活功能单元											

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Service Name 服务名称	SID (Hex)	Sub-function/ 子功能 (Hex)	Session/会话						Security Access[1]/ 安全访问		
			Default 默认		Programming 编程		Extended 扩展		LV1 等级 1	LV 2 等级 2	LV 3 等级 3
			Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址	Phys 物理寻址	Func 功能寻址			
RoutineControl	31	01/03	×	×	√	×	√	×	×	√	√
Upload download functional unit/上传下载功能单元											
RequestDownload	34	-	×	×	√	×	×	×	×	×	√
TransferData	36	-	×	×	√	×	×	×	×	×	×
RequestTransferExit	37	-	×	×	√	×	×	×	×	×	×

NOTE:
注：

1. The definition of the security access level can be found in chapter 错误！未找到引用源。 of this specification.
有关安全访问等级的定义可参见本规范中的第错误！未找到引用源。章节。
2. '√': Support; '×': Not support, '-': Not Applicable.(N/A)
"√"：支持；"×"：不支持；"-"不适用。
3. If the ReadDataByIdentifier (22 hex) service request message is a single frame, the functional addressing method shall be supported; otherwise, the functional addressing method is not required to be supported.
如果通过标识符读数据（22 hex）服务请求消息为单帧，则应（shall）支持功能寻址，否则，不需要支持功能寻址。
4. All the WriteDataByIdentifier (2E hex) service in all ECUs shall need the security access, and the security level needed by this service depends on each DID(see annex C)
所有 ECU 的通过标识符写数据（2E hex）服务均应（shall）需要安全访问，该服务所需的安全等级取决于具体 DID（见附录 C）。
5. the RoutineControl (31 hex) service need the security access, the security level needed by this service depends on each RID which is defined in this document.
RoutineControl (31 hex) 服务所需的安全访问等级，在本文附录中的 RID 中具体定义。

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9 Diagnostic service documentation structure

All the requirements about diagnostic service documentation structure, with which the tester and ECM shall comply, can be found in section 9 of the document *CN200S General Diagnostic Requirement Specification*.

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10 Diagnostic and communication management functional unit

10.1 DiagnosticSessionControl (10 hex)

10.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.1.2 Request message

Table 10-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Request Service Id	10
#2	DiagnosticSessionType=[DefaultSession ProgrammingSession ExtendedDiagnosticSession]	 01 02 03

10.1.3 Positive response message

Table 10-2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	DiagnosticSessionControl Response Service Id	50
#2	DiagnosticSessionType=[DefaultSession ProgrammingSession ExtendedDiagnosticSession]	 01 02 03
 #3 #4 #5 #6	sessionParameterRecord[]#1=[$P2_{CAN_ECU_max}$(high byte) $P2_{CAN_ECU_max}$(low byte) $P2^*_{CAN_ECU_max}$(high byte) $P2^*_{CAN_ECU_max}$(low byte)]	 00 32 01 F4

NOTE: The definition about parameter $P2_{CAN_ECU_max}$ and $P2^*_{CAN_ECU_max}$ can be found in chapter 8.2.2 of the document *CN200S General Diagnostic Requirement Specification* and ISO 15765-3. These two parameters in positive response message are the ones that are used by ECM internally, if the value of this two parameters which used by ECM are different from the definition in this document, it should be determined by SGMW and ECM Supplier together.

10.1.4 Supported negative response codes

Table 10-3 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

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10.2 ECUReset (11 hex)

10.2.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.2.2 Request message

Table 10-4 Request message definition

A_Data Byte	Parameter name	Hex value
#1	ECUReset Request Service Id	11
#2	resetType =[hardReset keyOffOnReset]	 01 02

10.2.3 Positive response message

Table 10-5 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ECUReset Response Id	51
#2	resetType =[hardReset keyOffOnReset]	 01 02

10.2.4 Supported negative response codes

Table 10-6 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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10.3 SecurityAccess (27 hex)

10.3.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

NOTE: This service, as well as MEC provides a protected mechanism to access the diagnostic services. The detailed definition about MEC can be found in chapter 16.1.3.2.5 of the document *CN200S General Diagnostic Requirement Specification*. The algorithm should be provided by SGMW.

NOTE: The MEC mechanism is only applicable to Level1 of security access service (27 hex).
注：MEC机制只适用于安全访问服务的安全等级Level1.

NOTE: The detailed requirements about SecurityAccess (27 hex) service can be found in chapter 10.3 of the document *CN200S General Diagnostic Requirement Specification.*

10.3.2 Request message

Table 10-7 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType=requestSeed	01/03/05

Table 10-8 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	SecurityAccess Request Service Id	27
#2	securityAccessType =sendKey	02/04/06
	securityKey[]=[	
#3	key #1(MSB)	00-FF
#4	key #2	:
#5	key #3	:
#6	key#4(LSB)]	00-FF

10.3.3 Positive response message

Table 10-9 Positive response message definition (sub-function = requestSeed)

A_Data Byte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType=requestSeed	01/03/05
	securitySeed[]=[	
#3	seed#1(MSB)	00-FF
#4	seed#2	:
#5	seed#3	:
#6	seed#4(LSB)]	00-FF

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Table 10-10 Positive response message definition (sub-function = sendKey)

A_Data Byte	Parameter Name	Hex value
#1	Security Access Response Service Id	67
#2	securityAccessType =sendKey	02/04/06

10.3.4 Supported negative response codes

Table 10-11 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**
24	**requestSequenceError**
35	**invalidKey**
36	**exceededNumberOfAttempts**
37	**requireTimeDelayNotExpired**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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10.4 CommunicationControl (28 hex)

N/A

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10.5 TesterPresent (3E hex)

10.5.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

10.5.2 Request message

Table 10-12 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	TesterPresent Request Service Id	3E
#2	zeroSubFunction=[zeroSubFunction (suppressPosRspMsgIndicationBit =0) zeroSubFunction (suppressPosRspMsgIndicationBit =1)]	 00 80

10.5.3 Positive response message

Table 10-13 Positive response message definition (sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =0))

A_Data Byte	Parameter Name	Hex value
#1	TesterPresent Response Service Id	7E
#2	zeroSubFunction= zeroSubFunction (suppressPosRspMsgIndicationBit =0)	00

sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of suppressPosRspMsgIndicationBit =1.

10.5.4 Supported negative response codes

Table 10-14 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

10.6 AccessTimingParameter (83 hex)

N/A

10.7 SecuredDataTransmission (84 hex)

N/A

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10.8 ControlDTCSetting (85 hex)

10.9 ResponseOnEvent (86 hex)

N/A

10.10 LinkControl (87 hex)

N/A

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10.11 ProgrammingMode (A5 hex)

N/A

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11 Data transmission functional unit

11.1 ReadDataByIdentifier (22 hex)

11.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

NOTE: The maximum number of dataIdentifier that can be simultaneously requested is one.

NOTE: The DID which used to identify the general data record for all the ECUs called vehicle-manufacturer-specific DID, or the definition about vehicle-manufacturer-specific DID can be found in chapter C.1 of Annex C. The DID which used to identify the individual data record for each ECU called ECU-supplier-specific DID, the definition about ECU-supplier-specific DID can be found in in chapter C.2 of Annex C.

11.1.2 Request message

Table 11-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	ReadDataByIdentifier Request Service Id	22
	dataIdentifier[]#1=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF

11.1.3 Positive response message

Table 11-2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ReadDataByIdentifier Response Service Id	62
	dataIdentifier[]#1=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
	dataRecord[]#1=[	
#4	data #1	00-FF
:	:	:
#(k-1)+4	data #k [a]]	00-FF
a. The number of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C.		

11.1.4 Supported negative response codes

Table 11-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the

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requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

11.2 ReadMemoryByAddress (23 hex)

N/A

11.3 ReadScalingDataByIdentifier (24 hex)

N/A

11.4 ReadDataByPeriodicIdentifier (2A hex)

N/A

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11.5 DynamicallyDefineDataIdentifier (2C hex)

11.6 WriteDataByIdentifier (2E hex)

11.6.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

11.6.2 Request message

Table 11-4 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	WriteDataByIdentifier Request Service Id	2E
	dataIdentifier[]=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF
	dataRecord[]=[	
#4	data #1	00-FF
:	:	:
#m+3	data #m[a]]	00-FF
a. The content of data# depends on specified dataIdentifier in chapter C.1 and chapter C.2 of Annex C.		

11.6.3 Positive response message

Table 11-5 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	WriteDataByIdentifier Response Service Id	6E
	dataIdentifier[]=[	
#2	byte #1(MSB)	00-FF
#3	byte #2(LSB)]	00-FF

11.6.4 Supported negative response codes

Table 11-6 Supported negative response codes

Hex	Description
33	**securityAccessDenied**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**
72	**generalProgrammingFailure**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification.*

11.7 WriteMemoryByAddress (3D hex)

N/A

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12 Stored data transmission functional unit

12.1 ClearDiagnosticInformation (14 hex)

12.1.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

12.1.2 Request message

Table 12-1 Request message definition

A_Data Byte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Request Service Id	14
	groupOfDTC[]=[	
#2	groupOfDTCHighByte	FF
#3	groupOfDTCMiddleByte	FF
#4	groupOfDTCLowByte]	FF

12.1.3 Positive response message

Table 12-2 Positive response message definition

A_Data Byte	Parameter Name	Hex value
#1	ClearDiagnosticInformation Response Service Id	54

12.1.4 Supported negative response codes

Table 12-3 Supported negative response codes

Hex	Description
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**
22	**conditionsNotCorrect**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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12.2 ReadDTCInformation (19 hex)

12.2.1 Service description

All general requirements about the service can be found in related section of *CN200S General Diagnostic Requirement Specification.*

12.2.2 Request message

Table 12-4 Request message definition (sub-function = reportNumberOfDTCByStatusMask)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=[reportNumberOfDTCByStatusMask reportDTCByStatusMask]	 01 02
#3	DTCStatusMask	00-FF

Table 12-5 Request message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
 #3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCSnapshotRecordNumber	FF[a]
a. A value of FF hex requests the ECU to report all stored DTCSnapshot data records at once.		

Table 12-6 Request message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Request Service Id	19
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
 #3 #4 #5	DTCMaskRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte]	 00-FF 00-FF 00-FF
#6	DTCExtendedDataRecordNumber	00-FF[a]
a. A value of FF hex requests the ECU to report all stored DTCExtendedData records in a single response message.		

12.2.3 Positive response message

Table 12-7 Positive response message definition (sub-function = reportNumberOfDTCByStatusMask)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportNumberOfDTCByStatusMask	01

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#3	DTCStatusAvailabilityMask	FF
#4	ISO 15031-6 DTCFormat	00
#5 #6	DTCCount[]=[DTCCountHighByte DTCCountLowByte]	00-FF 00-FF

Table 12-8 Positive response message definition (sub-function = reportDTCByStatusMask, reportSupportedDTC)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=[reportDTCByStatusMask]	02
#3	DTCStatusAvailabilityMask	FF
#4 #5 #6 #7 #8 #9 #10 #11 : #n-3 #n-2 #n-1 #n	DTCAndStatusRecord[] = [DTCHighByte #1 DTCMiddleByte #1 DTCLowByte #1 statusOfDTC #1 DTCHighByte #2 DTCMiddleByte #2 DTCLowByte #2 statusOfDTC #2 : DTCHighByte #m DTCMiddleByte #m DTCLowByte #m statusOfDTC #m]	00-FF 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF 00-FF : 00-FF 00-FF 00-FF 00-FF
NOTE: If reportType = reportSupportedDTC: 1. ECM should report the status of all the DTC's which is supported by ECM. 2. ECM should report the status at the time of receiving the requirement.		

Table 12-9 Positive response message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCSnapshotRecordByDTCNumber	04
#3 #4 #5 #6	DTCAndStatusRecord[]=[DTCHighByte DTCMiddleByte DTCLowByte statusOfDTC]	00-FF 00-FF 00-FF 00-FF
#7	DTCSnapshotRecordNumber#1	01
#8	DTCSnapshotRecordNumberOfIdentifiers#1	20

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	DTCSnapshotRecord[]#1=[	
#9	C_ERR_DTC_INFO DataIdentifier (MSB)	F0
#10	C_ERR_DTC_INFO DataIdentifier (LSB)	14
#11	C_ERR_DTC_INFO	00-FF
#12	Fuel system status DataIdentifier (MSB)	F0
#13	Fuel system status DataIdentifier (LSB)	15
#14	Fuel system status	00-FF
#15	Calculated load value DataIdentifier (MSB)	F0
#16	Calculated load value DataIdentifier (LSB)	16
#17	Calculated load value	00-FF
#18	Engine coolant temperature DataIdentifier (MSB)	F0
#19	Engine coolant temperature DataIdentifier (LSB)	17
#20	Engine coolant temperature	00-FF
#21	Long term fuel trim bank 1 DataIdentifier (MSB)	F0
#22	Long term fuel trim bank 1 DataIdentifier (LSB)	18
#23	Long term fuel trim bank 1 (High Byte)	00-FF
#24	Short term fuel trim bank 1 DataIdentifier (MSB)	F0
#25	Short term fuel trim bank 1 DataIdentifier (LSB)	19
#26	Short term fuel trim bank 1 (High Byte)	00-FF
#27	Intake manifold pressure DataIdentifier (MSB)	F0
#28	Intake manifold pressure DataIdentifier (LSB)	1A
#29	Intake manifold pressure	00-FF
#30	Vehicle speed sensor DataIdentifier (MSB)	F0
#31	Vehicle speed sensor DataIdentifier (LSB)	1B
#32	Vehicle speed sensor	00-FF
#33	Engine RPM DataIdentifier (MSB)	F0
#34	Engine RPM DataIdentifier (LSB)	1C
#35	Engine RPM (High Byte)	00-FF
#36	Engine RPM (Low Byte)	00-FF
#37	Relative throttle position DataIdentifier (MSB)	F0
#38	Relative throttle position DataIdentifier (LSB)	1D
#39	Relative throttle position	00-FF
#40	Applied ignition angle DataIdentifier (MSB)	F0
#41	Applied ignition angle DataIdentifier (LSB)	1E
#42	Applied ignition angle	00-FF
#43	Engine state DataIdentifier (MSB)	F0
#44	Engine state DataIdentifier (LSB)	1F
#45	Engine state	00-FF
#46	Time since engine start DataIdentifier (MSB)	F0
#47	Time since engine start DataIdentifier (LSB)	20
#48	Time since engine start	00-FF
#49	Fault status DataIdentifier (MSB)	F0
#50	Fault status DataIdentifier (LSB)	21
#51	Fault status (High Byte)	00-FF
#52	Fault status (Low Byte)	00-FF
#53	Number of diving cycles since DTC cleared DataIdentifier (MSB)	F0
#54	Number of diving cycles since DTC cleared DataIdentifier (LSB)	22
#55	Number of diving cycles since DTC cleared (High Byte)	00-FF
#56	Number of diving cycles since DTC cleared (Low Byte)	00-FF
#57	Distance since DTC cleared DataIdentifier (MSB)	F0

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#58	Distance since DTC cleared DataIdentifier (LSB)	23
#59	Distance since DTC cleared (High Byte)	00-FF
#60	Distance since DTC cleared (Low Byte)	00-FF
#61	1 DataIdentifier (MSB)	F0
#62	1 DataIdentifier (LSB)	24
#63	1	00-FF
#64	Environmental data 0 for general monitoring error DIDr (MSB)	F0
#65	Environmental data 0 for general monitoring error DID (LSB)	25
#66	Environmental data 0 for general monitoring error	00-FF
#67	Environmental data 1 for general monitoring error DID (MSB)	F0
#68	Environmental data 1 for general monitoring error DID(LSB)	26
#69	Environmental data 1 for general monitoring error	00-FF
#70	Environmental data 2 for general monitoring error DID (MSB)	F0
#71	Environmental data 2 for general monitoring error DID (LSB)	27
#72	Environmental data 2 for general monitoring error	00-FF
#73	Environmental data 3 for general monitoring error DID (MSB)	F0
#74	Environmental data 3 for general monitoring error DID (LSB)	28
#75	Environmental data 3 for general monitoring error	00-FF
#76	Fuel system status DataIdentifier (MSB)	F0
#77	Fuel system status DataIdentifier (LSB)	29
#78	Fuel system status	00-FF
#79	Calculated load value DataIdentifier (MSB)	F0
#80	Calculated load value DataIdentifier (LSB)	2A
#81	Calculated load value	00-FF
#82	Engine coolant temperature DataIdentifier (MSB)	F0
#83	Engine coolant temperature DataIdentifier (LSB)	2B
#84	Engine coolant temperature	00-FF
#85	Short term fuel trim bank 1 DataIdentifier (MSB)	F0
#86	Short term fuel trim bank 1 DataIdentifier (LSB)	2C
#87	Short term fuel trim bank 1 (High Byte)	00-FF
#88	Short term fuel trim bank 1 (Low Byte)	00-FF
#89	Long term fuel trim bank 1 DataIdentifier (MSB)	F0
#90	Long term fuel trim bank 1 DataIdentifier (LSB)	2D
#91	Long term fuel trim bank 1 (High Byte)	00-FF
#92	Long term fuel trim bank 1 (Low Byte)	00-FF
#93	Intake manifold pressure DataIdentifier (MSB)	F0
#94	Intake manifold pressure DataIdentifier (LSB)	2E
#95	Intake manifold pressure	00-FF
#96	Engine RPM DataIdentifier (MSB)	F0
#97	Engine RPM DataIdentifier (LSB)	2F
#98	Engine RPM (High Byte)	00-FF
#99	Engine RPM (Low Byte)	00-FF
#100	Vehicle speed sensor DataIdentifier (MSB)	F0
#101	Vehicle speed sensor DataIdentifier (LSB)	30
#102	Vehicle speed sensor	00-FF
#103	Location of oxygen sensors DataIdentifier (MSB)	F0
#104	Location of oxygen sensors DataIdentifier (LSB)	31
#105	Location of oxygen sensors	00-FF
#106	Short term secondary O2 sensor fuel trim bank 1 DID (MSB)	F0
#107	Short term secondary O2 sensor fuel trim bank 1 DID (LSB)	32

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#108	Short term secondary O2 sensor fuel trim bank 1	00-FF
#109	Long term secondary O2 sensor fuel trim bank 1 DID (MSB)	F0
#110	Long term secondary O2 sensor fuel trim bank 1 DID (LSB)	33
#111	Long term secondary O2 sensor fuel trim bank 1	00-FF

Table 12-10 Positive response message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_Data Byte	Parameter Name	Hex value
#1	ReadDTCInformation Positive Response Service Id	59
#2	reportType=reportDTCExtendedDataRecordByDTCNumber	06
	DTCAndStatusRecord[]=[	
#3	DTCHighByte	00-FF
#4	DTCMiddleByte	00-FF
#5	DTCLowByte	00-FF
#6	statusOfDTC]	00-FF
#7	DTCExtendedDataRecordNumber #1[a]	01
#8	Fault occurrence counter[a]	00-FF
#9	DTCExtendedDataRecordNumber #2[a]	02
#10	Fault pending counter[a]	FF
#11	DTCExtendedDataRecordNumber #3[a]	03
#12	DTC Aging counter[a]	00-FF
#13	DTCExtendedDataRecordNumber #4[a]	04
#14	DTC Aged counter[a]	FF

a. This part is the DTCExtendedDataRecord for all ECUs, detailed definitions of DTC extended data record can be found in chapter 12.2.3.2 of *CN200S General Diagnostic Requirement Specification*. Whether the parameter appeared or not and the number of these parameters that appeared depend on the specific service requested.

NOTE: For ECM(1.5L), only Fault occurrence counter and DTC Aging counter are supported, for Fault pending counter and DTC Aged counter that ECM(1.5L) do not supported, the default value should be 0xFF.

12.2.4 Supported negative response codes

Table 12-11 Supported negative response codes

Hex	Description
12	**subFunctionNotSupported**
31	**requestOutOfRange**
13	**incorrectMessageLengthOrInvalidFormat**

NOTE: In addition to the NRC which listed in the table above, ECM(1.5L) should comply with the requirement of the notes defined in chapter 9.4 of the document *CN200S General Diagnostic Requirement Specification*.

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13 InputOutput control functional unit

13.1 InputOutputControlByIdentifier (2F hex)

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14 Remote activation of routine functional unit

14.1 RoutineControl (31 hex)

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15 Upload download functional unit

15.1 RequestDownload (34 hex)

15.2 RequestUpload (35 hex)

N/A

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15.3 TransferData (36 hex)

15.4 RequestTransferExit (37 hex)

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16 ECU programming requirements and process

The detailed description about this chapter can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

NOTE: For CN200S vehicle, ECM need to realize the software download function, the detailed requirements about software download can be found in chapter 16 of the document *CN200S General Diagnostic Requirement Specification*.

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Annex A (normative) Global parameter definitions

The detailed description about this Annex can be found in Annex A of the document *CN200S General Diagnostic Requirement Specification*.

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Annex B (normative) Diagnostic and communication management functional unit data parameter definitions

The detailed description about this Annex can be found in Annex B of the document *CN200S General Diagnostic Requirement Specification*.

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Annex C (normative) Data transmission functional unit data parameter definitions

C.1 Vehicle-manufacturer-specific DID definitions

C.2 ECU-supplier-specific DID definitions

The following table lists the definition about ECU-supplier-specific DID. According to the requirements of the document *CN200S General Diagnostic Requirement Specification*, the range which ECM used is 0x0101~0x0FFF.

DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0200	Maximum Attainable Indicated Torque	√	x	2	Hex
0201	Desired Indicated Engine Torque	√	x	2	Hex
0202	Indicated Real Engine Torque	√	x	2	Hex
0203	Engine Temperature	√	x	1	Hex
0204	Engine Speed	√	x	2	Hex
0205	Time Counter At End Of Start	√	x	2	Hex
0206	Fuel Level	√	x	2	Hex
0207	Oil Temperature In Cylinder Head	√	x	1	Hex
0208	Vehicle Speed	√	x	1	Hex
0209	Ambient Pressure	√	x	2	Hex
020A	Battery Voltage	√	x	1	Hex
020B	Throttle Angle with Respect To Lower Mechanical Stop	√	x	1	Hex
020C	Filtered Rail Pressure Real Value (Absolute Pressure)	√	x	2	Hex
020D	Intake Air Temperature	√	x	1	Hex
020E	Ambient Air Temperature	√	x	1	Hex
020F	Pressure Upstream Throttle Valve	√	x	2	Hex
0210	Normalized Angle Acceleration Pedal	√	x	1	Hex
0211	Desired Pressure Upstream Throttle Valve	√	x	2	Hex
0212	Actual Exhaust Gas Pressure Downstream Of Exhaust Valve	√	x	2	Hex
0213	Exhaust Gas Temperature At Manifold Out Of Model	√	x	2	Hex
0214	Fuel System 1 Status	√	x	1	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0215	Fuel System 2 Status	√	x	1	Hex
0216	Relative Air Mass (Calc. Load Value) Acc. To Sae J1979	√	x	1	Hex
0217	Air Mass Flow	√	x	2	Hex
0218	Output Voltage Oxygen Sensor Downstream Catalyst	√	x	2	Hex
0219	Output Voltage Oxygen Sensor Downstream Catalyst Bank 2	√	x	2	Hex
021A	Ets-Path As Environmental Condition For Diagnosis Of Function Monitoring	√	x	2	Hex
021B	Torque Path In Function And Function Monitoring As Environmental Cond. Of Diag	√	x	2	Hex
021D	Model-Based Substitute Value For Engine Temperature Signal In Case Of Error	√	x	2	Hex
021E	Desired Throttle Angle W.R.T. To Lower Mechanical Stop	√	x	2	Hex
021F	Exhaust Temperature Downstream Catalyst (Model)	√	x	2	Hex
0221	Engaged Gear	√	x	1	Hex
0222	Real Ignition Angle	√	x	1	Hex
0223	Throttle Blade Position Dependent On Poti 1	√	x	2	Hex
0224	Throttle Blade Position Dependent On Poti 2	√	x	2	Hex
0225	Dlr For Dv-E: Sum Of The Pid-Components (Throttle Actuator Control Pwm)	√	x	2	Hex
0226	Diagnosis: 8 Bit Copy Of wdknlp_Was Ambient Value	√	x	2	Hex
0227	Sensor Voltage Poti 1 Of Throttle Actuator At (Lower) Mechanical Stop	√	x	2	Hex
0228	Intake Manifold Pressure (Absolute)	√	x	2	Hex
022B	Battery Voltage; Scanned Value Of Adc	√	x	2	Hex
022C	Angle: Intake Valve Opens Related To Lwot	√	x	2	Hex
022D	Desired Camshaft Angle Inlet Valve Opened	√	x	1	Hex
022F	Lambda Controller Output	√	x	2	Hex
0230	Lambda Controller Output Bank 2	√	x	2	Hex
0231	Multiplicative Correction Of The Mixture Adaptation	√	x	2	Hex
0232	Sensor Voltage From Throttle Potentiometer 1	√	x	2	Hex
0233	Sensor Voltage From Throttle Potentiometer 2	√	x	2	Hex
0234	Additive Correction Of The Mixture Adaptation	√	x	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0235	Actual Value Internal Res. Ri-Nernst Cell Lambda Sensor Downstream Cat	√	x	2	Hex
0236	Exhaust Gas Temperature In Front Of Pre-Catalyst Out Of Model	√	x	2	Hex
0237	Transfer Function (Mrad/Mkurbelwelle) From Transmission Control	√	x	2	Hex
0239	Interface For Mode $01+$02 Long Term Fuel Trim Bank 1 For Pid $07	√	x	2	Hex
023A	Injection Time	√	x	8	Hex
023C	Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #1	√	x	1	Hex
023D	Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #2	√	x	1	Hex
023E	Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #3	√	x	1	Hex
023F	Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #4	√	x	1	Hex
0240	Reference Level Knock Control, 16bit, Cylinder #1	√	x	2	Hex
0241	Reference Level Knock Control, 16bit, Cylinder #2	√	x	2	Hex
0242	Reference Level Knock Control, 16bit, Cylinder #3	√	x	2	Hex
0243	Reference Level Knock Control, 16bit, Cylinder #4	√	x	2	Hex
0244	Counter For Distance Travelled While Mil Is Activated	√	x	2	Hex
0245	Engine Temperature At Start	√	x	1	Hex
0246	Standardized Mass Flow From Canister Purge Control	√	x	2	Hex
0247	Basic Ignition Angle	√	x	1	Hex
0248	Fault Counter, Summary, Counts Emission Relevantmisfirings Of All Cylinders	√	x	2	Hex
0249	Number Of Misfire:Cylinder#1	√	x	2	Hex
024A	Number Of Misfire:Cylinder#2	√	x	2	Hex
024B	Number Of Misfire:Cylinder#3	√	x	2	Hex
024C	Number Of Misfire:Cylinder#4	√	x	2	Hex
024D	Area Of Misfire,Minimum Engine Speed	√	x	2	Hex
024E	Area Of Misfire,Maximum Engine Speed	√	x	2	Hex
024F	Area Of Misfirings, Minimum Load	√	x	1	Hex
0250	Cycle Time Forced Lambda Modulation	√	x	2	Hex
0251	Desired Torque Change From The Idle Speed Control (Pd-Part)	√	x	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0252	Idle Speed Control	√	x	2	Hex
0253	Canister Purge Solenoid	√	x	1	Hex
0254	Fuel Pump Relay	√	x	1	BIN
0255	Cooling Fan Relay	√	x	1	Hex
0257	Inlet Camshaft Control	√	x	1	Hex
0259	MIL	√	x	1	BIN
025A	SVS	√	x	1	BIN
025B	Output Duty Cycle For Canister Purge Valve	√	x	1	Hex
025D	Desired Throttle Position	√	x	2	Hex
025E	Duty Cycle To Control Power Stage Inlet Camshaft	√	x	2	Hex
0260	O2 Sensor Voltage Bank1, Sensor 1	√	x	2	Hex
0261	O2 Sensor Voltage Bank1, Sensor 2	√	x	2	Hex
0262	Actual Intake Manifold Pressure Sensor Voltage	√	x	2	Hex
0263	Delta Resistant Torque From Resistant Torque Adaptation	√	x	2	Hex
02F8	V_TPS_1	√	x	2	Hex
02F9	V_TPS_2	√	x	2	Hex
02FA	V_PVS_1	√	x	2	Hex
02FB	V_PVS_2	√	x	2	Hex
02FC	V_TCO	√	x	2	Hex
02FD	V_TIA	√	x	2	Hex
02FE	V_MAP	√	x	2	Hex
02FF	V_VB_BAS	√	x	2	Hex
0300	ACP	√	x	1	Hex
0301	V_ACP	√	x	2	Hex
0302	V_PSTE	√	x	2	Hex
0303	PSTE	√	x	2	Hex
0304	ALTPWM	√	x	1	Hex
0305	Status_1	√	x	1	BIN

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0306	Status_2	√	X	1	BIN
0307	Status_3	√	X	1	BIN
0308	Status_4	√	X	1	BIN
0309	Status_5	√	X	1	BIN
030A	LSHPWM_UP_1	√	X	1	Hex
030B	LSHPWM_DOWN_1	√	X	1	Hex
030C	VVTPWM_IN	√	X	2	Hex
030D	ECFPWM	√	X	1	Hex
030E	TI_1	√	X	2	Hex
030F	TI_2	√	X	2	Hex
0310	TI_3	√	X	2	Hex
0311	TI_4	√	X	2	Hex
0312	FAC_LAM_COR	√	X	2	Hex
0313	FAC_LAM_AD_OUT	√	X	2	Hex
0314	IGA_CYL_KNK[0]	√	X	1	Hex
0315	IGA_CYL_KNK[1]	√	X	1	Hex
0316	IGA_CYL_KNK[2]	√	X	1	Hex
0317	IGA_CYL_KNK[3]	√	X	1	Hex
0318	VP_TACE_SENS	√	X	2	Hex
0319	TACE	√	X	1	Hex
031A	TPS_AD_MMV_IS	√	X	8	Hex
031B	MAF_KGH	√	X	2	Hex
031C	MAF_FAC_ALTI_MMV	√	X	4	Hex
031D	SOI_0	√	X	2	Hex
031E	SOI_1	√	X	2	Hex
031F	SOI_2	√	X	2	Hex
0320	SOI_3	√	X	2	Hex
0321	PAT_INH_IV	√	X	1	Hex
0322	TI_LAM_1	√	X	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0323	TI_AD_ADD_MMV_1	√	x	2	Hex
0324	CL_MMV	√	x	2	Hex
0325	TD	√	x	2	Hex
0326	DIST_LW / DIST_HW	√	x	4	Hex
0327	CTR_ERR_VS	√	x	1	Hex
0328	TRT_ERR_VS	√	x	4	Hex
0329	STATE_IVVT	√	x	1	Hex
032A	STATE_IVVT_PWM_IN	√	x	1	Hex
032B	IMMO related information	√	x	1	BIN
032C	Counters of possible/actual stop-start event in the current drive cycle/life time	√	x	8	Hex
032D	Stop-start duration of the current driving cycle Total stop-start duration over life time	√	x	8	Hex
032E	EOL stop-start check	√	x	15	Hex&BIN
032F	Warm up reset counter	√	x	1	Hex
0330	Warm up reset type	√	x	1	Hex
0331	Environmental data 0 for general processor monitoring error	√	x	1	Hex
0332	Environmental data 1 for general processor monitoring error	√	x	1	Hex
0333	Environmental data 2 for general processor monitoring error	√	x	1	Hex
0334	Environmental data 3 for general processor monitoring error	√	x	1	Hex
03F8	CTR_SEG_AD_ER	√	x	1	Hex
03F9	FAC_AD_CEI_L[0]	√	x	4	Hex
03FA	FAC_AD_CEI_L[1]	√	x	4	Hex
03FB	FAC_AD_CEI_H[0]	√	x	4	Hex
03FC	FAC_AD_CEI_H[1]	√	x	4	Hex
03FD	PSN_EDGE_AD_CAM_IN[0][0]	√	x	2	Hex
03FE	PSN_EDGE_AD_CAM_IN[0][1]	√	x	2	Hex
03FF	PSN_EDGE_AD_CAM_IN[0][2]	√	x	2	Hex
0400	PSN_EDGE_AD_CAM_IN[0][3]	√	x	2	Hex
0401	PSN_EDGE_AD_CAM_IN[0][4]	√	x	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
0402	PSN_EDGE_AD_CAM_IN[0][5]	√	x	2	Hex
0403	PSN_EDGE_AD_CAM_IN[0][6]	√	x	2	Hex
0404	PSN_EDGE_AD_CAM_IN[0][7]	√	x	2	Hex
0405	PSN_EDGE_AD_CAM_IN[1][0]	√	x	2	Hex
0406	PSN_EDGE_AD_CAM_IN[1][1]	√	x	2	Hex
0407	PSN_EDGE_AD_CAM_IN[1][2]	√	x	2	Hex
0408	PSN_EDGE_AD_CAM_IN[1][3]	√	x	2	Hex
0409	PSN_EDGE_AD_CAM_IN[1][4]	√	x	2	Hex
040A	PSN_EDGE_AD_CAM_IN[1][5]	√	x	2	Hex
040B	PSN_EDGE_AD_CAM_IN[1][6]	√	x	2	Hex
040C	PSN_EDGE_AD_CAM_IN[1][7]	√	x	2	Hex
040D	PSN_EDGE_AD_CAM_EX[0][0]	√	x	2	Hex
040E	PSN_EDGE_AD_CAM_EX[0][1]	√	x	2	Hex
040F	TCO_STOP	√	x	1	Hex
0410	AR_RED_AD_ADD	√	x	2	Hex
0411	AR_RED_AD_ADD_MMV	√	x	2	Hex
0412	AR_RED_AD_FAC	√	x	2	Hex
0413	AR_RED_AD_FAC_COR	√	x	2	Hex
0414	AMP_AD	√	x	2	Hex
0415	PRS_AIR_MES_ADJ_OFS[0]	√	x	2	Hex
0416	PRS_AIR_MES_ADJ_OFS[1]	√	x	2	Hex
0417	PRS_AIR_MES_ADJ_OFS[2]	√	x	2	Hex
0418	FAC_AD_KNK	√	x	1	Hex
0419	Status_6	√	x	2	Hex
041A	T_DLY_I_AD_LAM_ADJ	√	x	2	Hex
041B	FAC_MFF_ADD_FAC_LAM_AD	√	x	2	Hex
041C	FAC_MPL_LAM_AD[0][0]	√	x	2	Hex
041D	FAC_MPL_LAM_AD[0][1]	√	x	2	Hex
041E	MFF_MPL_ADD_LAM_AD[0][0]	√	x	2	Hex

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DID (Hex)	Data Description	Supported by Service(Hex)		Data Length (Byte)	Data Type
		22	2E		
041F	MFF_MPL_ADD_LAM_AD[0][1]	√	x	4	Hex
0420	TQ_DIF_IS_AD_INTER	√	x	4	Hex
0421	TQ_DIF_IS_AD_ACC_CONV_INTER	√	x	4	Hex
0422	TQ_DIF_IS_AD_ACC_1_INTER	√	x	4	Hex
0423	TQ_DIF_IS_AD_CONV_INTER	√	x	1	Hex
0424	Status_7	√	x	1	BIN
0425	Status_8	√	x	1	BIN
0426	CONF_ENG	√	x	1	Hex
0427	ERR_GAS	√	x	2	Hex
0428	T_PEAK_IV_GAS_TI	√	x	2	Hex
0429	TI_ADD_DLY	√	x	1	Hex
042A	TGAS_L	√	x	1	Hex
042B	TGAS_H	√	x	2	Hex
042C	PGAS_L	√	x	2	Hex
042D	PGAS_H	√	x	1	Hex
042E	FTL_AFU	√	x	4	Hex
042F	TRT	√	x	4	Hex
0430	TRT_AFU	√	x	1	Hex
0431	Status_9	√	x	2	BIN
0432	MASS_GAS_PERC	√	x	2	Hex
0500	Cruise Status	√	x	1	Hex
0501	Cruise speed	√	x	2	Hex

NOTE: " √ ": Support without security access; " √ ": Support with security access (A superscript number 'n' after ' √ ' indicates that the number of the security level); "x": Not Supported.

C.2.1 DID 0x0200-Maximum Attainable Indicated Torque

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Maximum Attainable Indicated Torque	2	Nm	0.03125	8000…7FFFH	-1024…1023.96875

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C.2.2 DID 0x0201-Desired Indicated Engine Torque

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Desired Indicated Engine Torque	2	Nm	0.03125	8000…7FFFH	-1024…1023.96875

C.2.3 DID 0x0202-Indicated Real Engine Torque

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Indicated Real Engine Torque	2	Nm	0.03125	8000…7FFFH	-1024…1023.96875

C.2.4 DID 0x0203-Engine Temperature

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Engine Temperature	1	° C	0.75	0…FEH	-48…142.5

C.2.5 DID 0x0204-Engine Speed

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Engine Speed	2	rpm	1	0…1FE0H	0…8160

C.2.6 DID 0x0205-Time Counter At End Of Start

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Time Counter At End Of Start	2	S	0.1	0…FFFFH	0…6556.5

C.2.7 DID 0x0206-Fuel Level

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Fuel Level	1	%	0.390625	00-FF	0-99.609375

C.2.8 DID 0x0207-Oil Temperature in Cylinder Head

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Oil Temperature In Cylinder Head	1	° C	1	0…C8H	-40…160

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C.2.9 DID 0x0208-Vehicle Speed

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Vehicle Speed	1	km/h	1	0…FFH	0…255

C.2.10 DID 0x0209-Ambient Pressure

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ambient Pressure	2	hPa	0.08291752498664835	0…FFFFH	0…5434

C.2.11 DID 0x020A-Battery Voltage

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Battery Voltage	1	V	0.1015625	0…FFH	0…25.8984375

C.2.12 DID 0x020B-Throttle Angle with Respect to Lower Mechanical Stop

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Throttle Angle with Respect To Lower Mechanical Stop	1	° TPS	0.4668253677592627	0…FFH	0…119.040468778612

C.2.13 DID 0x020C-Filtered Rail Pressure Real Value (Absolute Pressure)

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Filtered Rail Pressure Real Value (Absolute Pressure)	2	hPa	5.306721599145495	0…FFFFH	0…347776

C.2.14 DID 0x020D-Intake Air Temperature

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Intake Air Temperature	1	° C	° C	0…FEH	-48…142.5

C.2.15 DID 0x020E-Ambient Air Temperature

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ambient Air Temperature	1	° C	0.75	0…FEH	-48…142.5

C.2.16 DID 0x020F-Pressure Upstream Throttle Valve

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Pressure Upstream Throttle Valve	2	hPa	0.08291752498664835	0…FFFFH	0…5434

C.2.17 DID 0x0210-Normalized Angle Acceleration Pedal

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits .
Normalized Angle Acceleration Pedal	1	%	0.390625	0…FFH	0…99.609375

C.2.18 DID 0x0211-Desired Pressure Upstream Throttle Valve

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits .
Desired Pressure Upstream Throttle Valve	2	hPa	0.08291752498664835	0…FFFFH	0…5434

C.2.19 DID 0x0212-Actual Exhaust Gas Pressure Downstream Of Exhaust Valve

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Actual Exhaust Gas Pressure Downstream Of Exhaust Valve	2	hPa	0.08291752498664835	0…FFFFH	0…5434

C.2.20 DID 0x0213-Exhaust Gas Temperature at Manifold Out Of Model

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Exhaust Gas Temperature At Manifold Out Of Model	2	° C	0.0625	0…7FFFH	-273.15…1774.7875

C.2.21 DID 0x0214-Fuel System 1 Status

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Fuel System 1 Status	1		1	0…FFH	0…255

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C.2.22 DID 0x0215-Fuel System 2 Status

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Fuel System 2 Status	1		1	0…FFH	0…255

C.2.23 DID 0x0216-Relative Air Mass (Calc. Load Value) Acc. To SAE J1979

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Relative Air Mass (Calc. Load Value) Acc. To Sae J1979	1	%	0.392156862745098	0…FFH	0…100

C.2.24 DID 0x0217-Air Mass Flow

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Air Mass Flow	2	mg/stk	0.02119478141451133	0…FFFFH	0…1389

C.2.25 DID 0x0218-Output Voltage Oxygen Sensor Downstream Catalyst

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Output Voltage Oxygen Sensor Downstream Catalyst	2	V	0.0048828125	0…3FFH	0…4.9951171875

C.2.26 DID 0x0219-Output Voltage Oxygen Sensor Downstream Catalyst Bank 2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Output Voltage Oxygen Sensor Downstream Catalyst Bank 2	2	V	0.0048828125	0…3FFH	0…4.9951171875

C.2.27 DID 0x021A-Ets-Path as Environmental Condition for Diagnosis of Function Monitoring

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ets-Path As Environmental Condition For Diagnosis Of Function Monitoring	2	mg/stk	0.04238956282902281	8000H…7FFFH	-1389.02119478142…1388.97880521859

C.2.28 DID 0x021B-Torque Path In Function And Function Monitoring As Environmental Cond. Of Diag

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Torque Path In Function And Function Monitoring As Environmental Cond. Of Diag	2	Nm	0.03125	0…7FFFH	0…1023.96875

C.2.29 DID 0x021D-Model-Based Substitute Value for Engine Temperature Signal In Case Of Error

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Model-Based Substitute Value For Engine Temperature Signal In Case Of Error	2	° C	0.0029296875	0…FE00H	-48…142.5

C.2.30 DID 0x021E-Desired Throttle Angle W.R.T. To Lower Mechanical Stop

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Desired Throttle Angle W.R.T. To Lower Mechanical Stop	2	Nm	0.03125	8000H…7FFFH	-1024…1023.96875

C.2.31 DID 0x021F-Exhaust Temperature Downstream Catalyst (Model)

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Exhaust Temperature Downstream Catalyst (Model)	2	° C	0.0625	0…7FF0H	0…2047

C.2.32 DID 0x0221-Engaged Gear

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Engaged Gear	1		1	0…FH	0…15

C.2.33 DID 0x0222-Real Ignition Angle

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Real Ignition Angle	1	° CRK	0.375	0…FFH	-35.625…60

C.2.34 DID 0x0223-Throttle Blade Position Dependent on Poti 1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Throttle Blade Position Dependent On Poti 1	2	° TPS	0.007294146 371238479	C001H… 3FFFH	-119.5…119.5

C.2.35 DID 0x0224-Throttle Blade Position Dependent on Poti 2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Throttle Blade Position Dependent On Poti 2	2	° TPS	0.007294146 371238479	C001H… 3FFFH	-119.5…119.5

C.2.36 DID 0x0225-Dlr for Dv-E: Sum Of The Pid-Components (Throttle Actuator Control Pwm)

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Dlr For Dv-E: Sum Of The Pid-Components (Throttle Actuator Control Pwm)	2	%	0.003051757 8125	8000H…7 FFFH	-100…99.9969 482421875

C.2.37 DID 0x0226-Diagnosis: 8 Bit Copy Of wdknlp_Was Ambient Value

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Diagnosis: 8 Bit Copy Of wdknlp_Was Ambient Value	2	° TPS	0.007294146371238479	0…3FFH	0…119.5

C.2.38 DID 0x0227-Sensor Voltage Poti 1 of Throttle Actuator at (Lower) Mechanical Stop

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Sensor Voltage Poti 1 Of Throttle Actuator At (Lower) Mechanical Stop	2	V	0.0001525878906250001	8000H…7FFFH	-5…4.99984741210938

C.2.39 DID 0x0228-Intake Manifold Pressure (Absolute)

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Intake Manifold Pressure (Absolute)	2	hPa	0.08291752498664835	0…FFFFH	0…5434

C.2.40 DID 0x022B-Battery Voltage; Scanned Value of Adc

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Battery Voltage; Scanned Value Of Adc	2	V	0.0006256103515617609	0…FFFFH	0…40.9993743896

C.2.41 DID 0x022C-Angle: Intake Valve Opens Related To Lwot

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Angle: Intake Valve Opens Related To Lwot	2	° CRK	0.0078125	8000H…7FFFH	-256…255.9921875

C.2.42 DID 0x022D-Desired Camshaft Angle Inlet Valve Opened

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Desired Camshaft Angle Inlet Valve Opened	1	° CRK	0.5	0…FFH	0…127.5

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C.2.43 DID 0x022F-Lambda Controller Output

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Lambda Controller Output	2	%	0.001525878906250001	8000H…7FFFH	-50…49.9984741210938

C.2.44 DID 0x0230-Lambda Controller Output Bank 2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Lambda Controller Output Bank 2	2	%	0.001525878906250001	8000H…7FFFH	-50…49.9984741210938

C.2.45 DID 0x0231-Multiplicative Correction of the Mixture Adaptation

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Multiplicative Correction Of The Mixture Adaptation	2 * NC_CBK_EX_NR * NC_NR_LAM_AD	%	0.001525878906250001	8000H…7FFFH	-50…49.9984741210938

C.2.46 DID 0x0232-Sensor Voltage from Throttle Potentiometer 1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Sensor Voltage From Throttle Potentiometer 1	2	V	0.0001525878906250001	8000H…7FFFH	-5…4.99984741210938

C.2.47 DID 0x0233-Sensor Voltage from Throttle Potentiometer 2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Sensor Voltage From Throttle Potentiometer 2	2	V	0.0001525878906250001	8000H…7FFFH	-5…4.99984741210938

C.2.48 DID 0x0234-Additive Correction of the Mixture Adaptation

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Additive Correction Of The Mixture Adaptation	2	mg/stk	0.02119478141451133	8000H…7FFFH	-694.510597390707…694.4894026092931

C.2.49 DID 0x0235-Actual Value Internal Res. Ri-Nernst Cell Lambda Sensor

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Downstream Cat

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Actual Value Internal Res. Ri-Nernst Cell Lambda Sensor Downstream Cat	2	Ohm	1	0…FFFFH	0…65535

C.2.50 DID 0x0236-Exhaust Gas Temperature In Front Of Pre-Catalyst Out Of Model

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits .
Exhaust Gas Temperature In Front Of Pre-Catalyst Out Of Model	2	° C	0.0625	0…7FFFH	-273.15…1774.7875

C.2.51 DID 0x0237-Transfer Function (Mrad/Mkurbelwelle) From Transmission Control

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Transfer Function (Mrad/Mkurbelwelle) From Transmission Control	2	rpm/(km/h)	0.007812499885557336	0…FFFFH	0…511.99218

C.2.52 DID 0x0239-Interface for Mode $01+$02 Long Term Fuel Trim Bank 1 for Pid $07

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Interface For Mode $01+$02 Long Term Fuel Trim Bank 1 For Pid $07	2	%	0.001525878906250001	8000H…7FFFH	-50…49.9984741210938

C.2.53 DID 0x023A-Injection Time

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Injection Time	8	Ms	0.004	0…FFFFH	0…262.14

C.2.54 DID 0x023C-Ignition-Timing Retardation with Retardation for Dynamics, Cylinder #1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #1	1	° CRK	0.375	0..FFH	-35.625…60

C.2.55 DID 0x023D-Ignition-Timing Retardation with Retardation for Dynamics, Cylinder #2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #2	1	° CRK	0.375	0..FFH	-35.625…60

C.2.56 DID 0x023E-Ignition-Timing Retardation with Retardation for Dynamics, Cylinder #3

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #3	1	° CRK	0.375	0..FFH	-35.625…60

C.2.57 DID 0x023F-Ignition-Timing Retardation with Retardation for Dynamics, Cylinder #4

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Ignition-Timing Retardation with Retardation For Dynamics, Cylinder #4	1	° CRK	0.375	0..FFH	-35.625…60

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C.2.58 DID 0x0240-Reference Level Knock Control, 16bit, Cylinder #1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Reference Level Knock Control, 16bit, Cylinder #1	2	V	7.629394537 270161e-005	0…FFFFH	0…4.9999 2371

C.2.59 DID 0x0241-Reference Level Knock Control, 16bit, Cylinder #2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Reference Level Knock Control, 16bit, Cylinder #2	2	V	7.629394537 270161e-005	0…FFFFH	0…4.99992371

C.2.60 DID 0x0242-Reference Level Knock Control, 16bit, Cylinder #3

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Reference Level Knock Control, 16bit, Cylinder #3	2	V	7.629394537 270161e-005	0…FFFFH	0…4.9999 2371

C.2.61 DID 0x0243-Reference Level Knock Control, 16bit, Cylinder #4

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Reference Level Knock Control, 16bit, Cylinder #4	2	V	7.629394537 270161e-005	0…FFFFH	0…4.9999 2371

C.2.62 DID 0x0244-Counter for Distance Travelled While Mil Is Activated

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Counter For Distance Travelled While Mil Is Activated	2	Km	1	0…FFFFH	0…65535

C.2.63 DID 0x0245-Engine Temperature at Start

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Engine Temperature At Start	1	° C	0.75	0…FEH	-48…142.5

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C.2.64 DID 0x0246-Standardized Mass Flow from Canister Purge Control

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Standardized Mass Flow From Canister Purge Control	2	mg/stk	0.021194781 41451133	0…FFFFH	0…1389

C.2.65 DID 0x0247-Basic Ignition Angle

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Basic Ignition Angle	1	° CRK	0.375	0…FFH	-35.625…60

C.2.66 DID 0x0248-Fault Counter, Summary, Counts Emission Relevant misfirings of All Cylinders

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Fault Counter, Summary, Counts Emission Relevantmisfirings Of All Cylinders	2		1	0…FFFFH	0…65535

C.2.67 DID 0x0249-Number Of Misfire:Cylinder#1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Number Of Misfire: Cylinder#1	2		1	0…FFFFH	0…65535

C.2.68 DID 0x024A-Number Of Misfire:Cylinder#2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Number Of Misfire: Cylinder#2	2		1	0…FFFFH	0…65535

C.2.69 DID 0x024B-Number Of Misfire:Cylinder#3

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Number Of Misfire: Cylinder#3	2		1	0…FFFFH	0…65535

C.2.70 DID 0x024C-Number Of Misfire:Cylinder#4

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Number Of Misfire: Cylinder#4	2		1	0…FFFFH	0…65535

C.2.71 DID 0x024D-Area Of Misfire, Minimum Engine Speed

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Area Of Misfire, Minimum Engine Speed	2	Rpm	1	0…1FE0H	0…8160

C.2.72 DID 0x024E-Area Of Misfire, Maximum Engine Speed

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Area Of Misfire, Maximum Engine Speed	2	Rpm	1	0…1FE0H	0…8160

C.2.73 DID 0x024F-Area Of Misfirings, Minimum Load

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Area Of Misfirings, Minimum Load	1	%	0.0030517578125	0…7FFFH	0…99.9969482421875

C.2.74 DID 0x0250-Cycle Time Forced Lambda Modulation

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Cycle Time Forced Lambda Modulation	2	0.01953125	0.01953125	0…FFFFH	0…1279.98046875

C.2.75 DID 0x0251-Desired Torque Change From the Idle Speed Control (Pd-Part)

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Desired Torque Change From The Idle Speed Control (Pd-Part)	2	Nm	0.03125	8000H…7FFFH	-1024…1023.96875

C.2.76 DID 0x0252-Idle Speed Control

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Idle Speed Control	2	rpm	1	0…1FE0H	0…8160

C.2.77 DID 0x0253-Canister Purge Solenoid

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Canister Purge Solenoid	1	-	1	0…FFH	0…255

C.2.78 DID 0x0254-Fuel Pump Relay

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Fuel Pump Relay	1		1	0…1H	0…1

C.2.79 DID 0x0255-Cooling Fan Relay

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Cooling Fan Relay	1		1	0…FFH	0…255

C.2.80 DID 0x0257-Inlet Camshaft Control

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Inlet Camshaft Control	1		1	0…3H	0…3

C.2.81 DID 0x0259-MIL

Byte No.	Description	Data Type	Value (Hex)	
data #1	MIL	BIN	bit7	Unused
			bit6	Unused

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			bit5	Unused
			bit4	Unused
			bit3	Unused
			bit2	Unused
			bit1	Unused
			bit0	0:OFF;1:0N

C.2.82 DID 0x025A-SVS

Byte No.	Description	Data Type	Value (Hex)	
data #1	SVS	BIN	bit7	Unused
			bit6	Unused
			bit5	Unused
			bit4	Unused
			bit3	Unused
			bit2	Unused
			bit1	Unused
			bit0	0:OFF;1:0N

C.2.83 DID 0x025B-Output Duty Cycle for Canister Purge Valve

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Output Duty Cycle For Canister Purge Valve	1	%	0.390625	0..FFH	0…99.609375

C.2.84 DID 0x025D-Desired Throttle Position

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Desired Throttle Position	2	° TPS	0.007294146371238479	0…3FFH	0…119.5

C.2.85 DID 0x025E-Duty Cycle to Control Power Stage Inlet Camshaft

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Duty Cycle To Control Power Stage Inlet Camshaft	2	%	0.006103515625	0…4000H	0…100

C.2.86 DID 0x0260-O_2 Sensor Voltage Bank1, Sensor 1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
O2 Sensor Voltage Bank1, Sensor 1	2	V	0.0048828125	0…3FFH	0…4.9951171875

C.2.87 DID 0x0261-O_2 Sensor Voltage Bank1, Sensor 2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
O2 Sensor Voltage Bank1, Sensor 2	2	V	0.0048828125	0…3FFH	0…4.9951171875

C.2.88 DID 0x0262-Actual Intake Manifold Pressure Sensor Voltage

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Actual Intake Manifold Pressure Sensor Voltage	2	V	0.0001525878906250002	0…7FFFH	0…4.99984741210938

C.2.89 DID 0x0263-Delta Resistant Torque from Resistant Torque Adaptation

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Delta Resistant Torque From Resistant Torque Adaptation	2	Nm	0.03125	8000H…7FFFH	-1024…1023.96875

C.2.90 DID 0x02F8-V_TPS_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_TPS_1	2	V	0.0001526	8000H…7FFFH	-5…4.99985

C.2.91 DID 0x02F9-V_TPS_2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_TPS_2	2	V	0.0001526	8000H…7FFFH	-5…4.99985

C.2.92 DID 0x02FA-V_PVS_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_PVS_1	2	V	5/1024	0 ... 3FFH	0…4.9952

C.2.93 DID 0x02FB-V_PVS_2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_PVS_2	2	V	5/1024	0 ... 3FFH	0…4.9952

C.2.94 DID 0x02FC-V_TCO

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_TCO	2	V	0.000152588	0...7FFFH	0…4.99984741

C.2.95 DID 0x02FD-V_TIA

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_TIA	2	V	0.000152588	0...7FFFH	0…4.999847

C.2.96 DID 0x02FE-V_MAP

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_MAP	2	V	0.000152588	0...7FFFH	0…4.999847

C.2.97 DID 0x02FF-V_VB_BAS

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_VB_BAS	2	V	6.2561E- 4	0...FFFFH	0…40.9993744

C.2.98 DID 0x0300-ACP

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
ACP	1	kPa	12.156863	0...FFH	0…3100

C.2.99 DID 0x0301-V_ACP

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_ACP	2	V	0.1526e-3	0...7FFFH	0…4.99984

C.2.100 DID 0x0302-V_PSTE

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
V_PSTE	2	V	0.1526e-3	0...7FFFH	0…4.99984

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C.2.101 DID 0x0303-PSTE

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSTE	2	[bar]	0.25	0...258H	0…150

C.2.102 DID 0x0304-ALTPWM

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
ALTPWM	1	[%]	0.390625	0...FFH	0…99.60937

C.2.103 DID 0x0305-Status_1

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_1	BIN	bit7	LV_ACIN
			bit6	Unused
			bit5	Unused
			bit4	Unused
			bit3	Unused
			bit2	LV_BTS
			bit1	LV_BLS
			bit0	LV_CS

C.2.104 DID 0x0306-Status_2

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_2	BIN	bit7	LV_VIM
			bit6	unused
			bit5	LV_RLY_ST
			bit4	LV_RLY_EFP
			bit3	LV_RLY_MAIN
			bit2	STATE_CFA bit 2
			bit1	STATE_CFA bit 1
			bit0	LV_ACCRLY_OUT

C.2.105 DID 0x0307-Status_3

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_3	BIN	bit7	unused
			bit6	unused
			bit5	LV_LAM_LSCL
			bit4	unused
			bit3	unused
			bit2	unused
			bit1	LV_MIL

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			bit0	unused

C.2.106 DID 0x0308-Status_4

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_4	BIN	bit7	unused
			bit6	unused
			bit5	unused
			bit4	LV_PRS_SWT
			bit3	LV_PRS_ACC
			bit2	LV_RLY_ACCOUT
			bit1	LV_RLY_ACCOUT_CTRL
			bit0	LV_ACIN

C.2.107 DID 0x0309 -Status_5

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_5	BIN	bit7	unused
			bit6	unused
			bit5	unused
			bit4	unused
			bit3	unused
			bit2	unused
			bit1	unused
			bit0	LV_KNK

C.2.108 DID 0x030A-LSHPWM_UP_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
LSHPWM_UP_1	1	%	0.390625	0...FFH	0…99.60937

C.2.109 DID 0x030B-LSHPWM_DOWN_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
LSHPWM_DOWN_1	1	%	0.390625	0...FFH	0…99.60937

C.2.110 DID 0x030C-VVTPWM_IN

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
VVTPWM_IN	2	%	0.0061035156	0... 4000h	0…100

C.2.111 DID 0x030D-ECFPWM

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
ECFPWM	1	%	0.392	0... FFH	0…100

C.2.112 DID 0x030E-TI_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_1	2	Ms	0.004	0...FFFFH	0…262.14

C.2.113 DID 0x030F-TI_2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_2	2	Ms	0.004	0...FFFFH	0…262.14

C.2.114 DID 0x0310-TI_3

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_3	2	Ms	0.004	0...FFFFH	0…262.14

C.2.115 DID 0x0311-TI_4

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_4	2	Ms	0.004	0...FFFFH	0…262.14

C.2.116 DID 0x0312-FAC_LAM_COR

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_LAM_COR	2	%	0.00152588	8000... 7FFFH	-50…49.9984741211

C.2.117 DID 0x0313-FAC_LAM_AD_OUT

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_LAM_AD_OUT	2	%	0.00152588	8000... 7FFFH	-50…49.9984741211

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C.2.118 DID 0x0314-IGA_CYL_KNK[0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
IGA_CYL_KNK[0]	1	° CRK	0.375	0... 80H	-48…0

C.2.119 DID 0x0315-IGA_CYL_KNK[1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
IGA_CYL_KNK[1]	1	° CRK	0.375	0... 80H	-48…0

C.2.120 DID 0x0316-IGA_CYL_KNK[2]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
IGA_CYL_KNK[2]	1	° CRK	0.375	0... 80H	-48…0

C.2.121 DID 0x0317-IGA_CYL_KNK[3]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
IGA_CYL_KNK[3]	1	° CRK	0.375	0... 80H	-48…0

C.2.122 DID 0x0318-VP_TACE_SENS

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
VP_TACE_SENS	2	V	0.000152588	0...7FFFH	0…4.99984741

C.2.123 DID 0x0319-TACE

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TACE	1	C°	0.75	0... FEH	-48…142.5

C.2.124 DID 0x031A-TPS_AD_MMV_IS

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TPS_AD_MMV_IS	8	V	0.00015258	8000…7FFFH	-5…4.9998474

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C.2.125 DID 0x031B-MAF_KGH

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
MAF_KGH	2	kg/h	0.03125	0…FFFFH	0…2047.96875

C.2.126 DID 0x031C-MAF_FAC_ALTI_MMV

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
MAF_FAC_ALTI_MMV	4		3.0517578125e-5	0…FFFFH	0…1.999969

C.2.127 DID 0x031D-SOI_0

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
SOI_0	2	° CRK	0.375	0…780H	0…720

C.2.128 DID 0x031E-SOI_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
SOI_1	2	° CRK	0.375	0…780H	0…720

C.2.129 DID 0x031F-SOI_2

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
SOI_2	2	° CRK	0.375	0…780H	0…720

C.2.130 DID 0x0320-SOI_3

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
SOI_3	2	° CRK	0.375	0…780H	0…720

C.2.131 DID 0x0321-PAT_INH_IV

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PAT_INH_IV	1		1	0…FFH	0…255

C.2.132 DID 0x0322-TI_LAM_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_LAM_1	2	%	0.001525878	8000…7FFFH	-50…49.998474

C.2.133 DID 0x0323-TI_AD_ADD_MMV_1

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_AD_ADD_MMV_1	2	mg/stk	0.021194781	8000…7FFFH	-694.510597…694.489403

C.2.134 DID 0x0324-CL_MMV

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
CL_MMV	2		3.0517578125e-5	0…FFFFH	0…1.999969

C.2.135 DID 0x0325-TD

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TD	2	Ms	0.004	0…FFFFH	0…262.14

C.2.136 DID 0x0326-DIST_LW / DIST_H

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
DIST_LW / DIST_HW	4	Km	0.1	0…FFFFFFFFH	0…429496729.5

C.2.137 DID 0x0327-CTR_ERR_VS

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
CTR_ERR_VS	1		1	0…FFH	0…255

C.2.138 DID 0x0328-TRT_ERR_VS

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TRT_ERR_VS	4	Hr	2.7777777e-5	0…FFFFFFFFH	0…119304.6471

C.2.139 DID 0x0329-STATE_IVVT

Byte No.	Description	Data Type	Value (Hex)	
data #1	STATE_IVVT	Hex	0	PASSIVE
			1	READY
			2	ACT_ADAPT
			3	ENABLE
			4	LIMP_HOME

C.2.140 DID 0x032A-STATE_IVVT_PWM_IN

Byte No.	Description	Data Type	Value (Hex)	
data #1	STATE_IVVT_PWM_IN	Hex	0	HLD
			1	ACT
			2	PAS

C.2.141 DID 0x032B-IMMO related information

Byte No.	Description	Data Type	Value (Hex)	
data #1	IMMO related information	BIN	bit7	Unused
			bit6	Unused
			bit5	Unused
			bit4	Unused
			bit3	Unused
			bit2	Configuration Switch to invert AC Relay and Fuel output for Immobilizer Configuration (0: without immobilizer

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				wiring; 1: with immobilizer wiring)
			bit1	Configuration Switch to select or de-select Immobilizer (0: immobilizer activated; 1: immobilizer deactivated)
			bit0	True when Secret Key is programmed in ECM

C.2.142 DID 0x032C-Counters of possible/actual stop-start event in the current drive cycle/life time

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Counters of possible/actual stop-start event in the current drive cycle/life time	8		1	0…FFFFH	0…65535

C.2.143 DID 0x032D-Stop-start duration of the current driving cycle Total stop-start duration over life time

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Stop-start duration of the current driving cycle Total stop-start duration over life time	8	S	0.01	0… FFFFFFFFH	0…42949672.95

C.2.144 DID 0x032E-EOL stop-start check

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
EOL stop-start check	4	-	1	0	0	4294967295
	4	-	1	0	0	4294967295
	2	-	1	0	0	65535
	2	-	1	0	0	65535
	2	-	1	0	0	65535

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Data Name	#of bytes		Unit	Scaling	Offset	Min.	Max.
	1	bit7	-	1	0	Unused	Unused
		bit6	-			Unused	Unused
		bit5	-			Unused	Unused
		bit4	-			Unused	Unused
		bit3	-			Unused	Unused
		bit2	-			Unused	Unused
		bit1	-			Unused	Unused
		bit0	-			0	1

C.2.145 DID 0x032F-Warm up reset counter

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Warm up reset counter	1		1	0…FFH	0…255

C.2.146 DID 0x0330-Warm up reset type

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Warm up reset type	1	-	1	0…FFH	0…255

C.2.147 DID 0x0331-Environmental data 0 for general processor monitoring error

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Environmental data 0 for general processor monitoring error	1	-	1	0…FFH	0…255

C.2.148 DID 0x0332-Environmental data 1 for general processor monitoring error

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Environmental data 1 for general processor monitoring error	1	-	1	0…FFH	0…255

C.2.149 DID 0x0333-Environmental data 2 for general processor monitoring error

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Environmental data 2 for general processor monitoring error	1	-	1	0…FFH	0…255

C.2.150 DID 0x0334-Environmental data 3 for general processor monitoring error

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Environmental data 3 for general processor monitoring error	1	-	1	0…FFH	0…255

C.2.151 DID 0x03F8-CTR_SEG_AD_ER

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
CTR_SEG_AD_ER	1	-	1	0…FFH	0…255

C.2.152 DID 0x03F9-FAC_AD_CEI_L[0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_AD_CEI_L[0]	4	Nm	15.2587e-6	80000000... 7FFFFFFFH	-32768…32.7679999847e3

C.2.153 DID 0x03FA-FAC_AD_CEI_L[1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_AD_CEI_L[1]	4	Nm	15.2587e-6	80000000... 7FFFFFFFH	-32768…32.7679999847e3

C.2.154 DID 0x03FB-FAC_AD_CEI_H[0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_AD_CEI_H[0]	4	Nm	15.2587e-6	80000000... 7FFFFFFFH	-32768…32.7679999847e3

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C.2.155 DID 0x03FC-FAC_AD_CEI_H[1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_AD_CEI_H[1]	4	Nm	15.2587e-6	80000000... 7FFFFFFFH	-32768…32.7679998847e3

C.2.156 DID 0x03FD-PSN_EDGE_AD_CAM_IN[0][0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][0]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.157 DID 0x03FE-PSN_EDGE_AD_CAM_IN[0][1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][1]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.158 DID 0x03FF-PSN_EDGE_AD_CAM_IN[0][2]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][2]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.159 DID 0x0400-PSN_EDGE_AD_CAM_IN[0][3]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][3]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.160 DID 0x0401-PSN_EDGE_AD_CAM_IN[0][4]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][4]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

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C.2.161 DID 0x0402-PSN_EDGE_AD_CAM_IN[0][5]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][5]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.162 DID 0x0403-PSN_EDGE_AD_CAM_IN[0][6]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][6]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.163 DID 0x0404-PSN_EDGE_AD_CAM_IN[0][7]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[0][7]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.164 DID 0x0405-PSN_EDGE_AD_CAM_IN[1][0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][0]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.165 DID 0x0406-PSN_EDGE_AD_CAM_IN[1][1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][1]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.166 DID 0x0407-PSN_EDGE_AD_CAM_IN[1][2]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][2]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.167 DID 0x0408-PSN_EDGE_AD_CAM_IN[1][3]

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][3]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.168 DID 0x0409-PSN_EDGE_AD_CAM_IN[1][4]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][4]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.169 DID 0x040A-PSN_EDGE_AD_CAM_IN[1][5]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][5]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.170 DID 0x040B-PSN_EDGE_AD_CAM_IN[1][6]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][6]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.171 DID 0x040C-PSN_EDGE_AD_CAM_IN[1][7]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_IN[1][7]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.172 DID 0x040D-PSN_EDGE_AD_CAM_EX[0][0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_EX[0][0]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.173 DID 0x040E-PSN_EDGE_AD_CAM_EX[0][1]

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PSN_EDGE_AD_CAM_EX[0][1]	2	° CRK	0.0625	0... 3FFH	-32…31.9375

C.2.174 DID 0x040F-TCO_STOP

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TCO_STOP	1	° C	0.75	0...FEh	-48…142.5

C.2.175 DID 0x0410-AR_RED_AD_ADD

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
AR_RED_AD_ADD	2	cm²	894.07e-6	8000...7FFFh	-29.296875… 29.295980

C.2.176 DID 0x0411-AR_RED_AD_ADD_MMV

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
AR_RED_AD_ADD_MMV	2	cm²	894.07e-6	8000...7FFFh	-29.296875… 29.295980

C.2.177 DID 0x0412-AR_RED_AD_FAC

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
AR_RED_AD_FAC	2	%	1.52588e-3	8000...7FFFh	-50…49.998474

C.2.178 DID 0x0413-AR_RED_AD_FAC_COR

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
AR_RED_AD_FAC_COR	2	%	1.52588e-3	8000...7FFFh	-50…49.998474

C.2.179 DID 0x0414-AMP_AD

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
AMP_AD	2	hPa	0.0829175	0...FFFFh	0…5434

C.2.180 DID 0x0415-PRS_AIR_MES_ADJ_OFS[0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PRS_AIR_MES_ADJ_OFS[0]	2	hPa	0.0829175	8000...7FFFh	-2717.041459…2716.958541

C.2.181 DID 0x0416-PRS_AIR_MES_ADJ_OFS[1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PRS_AIR_MES_ADJ_OFS[1]	2	hPa	0.0829175	8000...7FFFh	-2717.041459…2716.958541

C.2.182 DID 0x0417-PRS_AIR_MES_ADJ_OFS[2]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PRS_AIR_MES_ADJ_OFS[2]	2	hPa	0.0829175	8000...7FFFh	-2717.041459…2716.958541

C.2.183 DID 0x0418-FAC_AD_KNK

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_AD_KNK	1	-	0.00390625	0...FFh	0…0.99609375

C.2.184 DID 0x0419-Status_6

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Status_6	1	℃RK	0.0375	0...7Fh	0…47.625

C.2.185 DID 0x041A-T_DLY_I_AD_LAM_AD

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
T_DLY_I_AD_LAM_ADJ	2	Ms	0.0195313	8000...7FFFh	-640…639.98046875

C.2.186 DID 0x041B-FAC_MFF_ADD_FAC_LAM_AD

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_MFF_ADD_FAC_LAM_AD	2	%	3.05176e-3	8000...7FFFh	-100…99.9969482422

C.2.187 DID 0x041C-FAC_MPL_LAM_AD[0][0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_MPL_LAM_AD[0][0]	2	%	1.52588e-3	8000...7FFFh	-50…49.99847412

C.2.188 DID 0x041D-FAC_MPL_LAM_AD[0][1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FAC_MPL_LAM_AD[0][1]	2	%	1.52588e-3	8000...7FFFh	-50…49.99847412

C.2.189 DID 0x041E-MFF_MPL_ADD_LAM_AD[0][0]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
MFF_MPL_ADD_LAM_AD [0][0]	2	mg/stk	0.0211948	8000...7FFFh	-694.510597391…694.489402609

C.2.190 DID 0x041F-MFF_MPL_ADD_LAM_AD[0][1]

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
MFF_MPL_ADD_LAM_AD [0][1]	2	mg/stk	0.0211948	8000...7FFFh	-694.510597391… 694.489402609

C.2.191 DID 0x0420-TQ_DIF_IS_AD_INTER

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TQ_DIF_IS_AD_INTER	4	Nm	476.837e-9	80000000... 7FFFFFFFh	-1024…1023. 999999

C.2.192 DID 0x0421-TQ_DIF_IS_AD_ACC_CONV_INTER

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TQ_DIF_IS_AD_ACC_C ONV_INTER	4	Nm	476.837e-9	80000000... 7FFFFFFFh	-1024…1023. 999999

C.2.193 DID 0x0422-TQ_DIF_IS_AD_ACC_1_INTER

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TQ_DIF_IS_AD_ACC_1 _INTER	4	Nm	476.837e-9	80000000... 7FFFFFFFh	-1024…1023. 999999

C.2.194 DID 0x0423-TQ_DIF_IS_AD_CONV_INTER

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TQ_DIF_IS_AD_CONV _INTER	4	Nm	476.837e-9	80000000... 7FFFFFFFh	-1024…1023. 999999

C.2.195 DID 0x0424-Status_7

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_7	BIN	bit7	unused
			bit6	unused
			bit5	unused

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			bit4	
			bit3	Bi-fuel Switch in Progress
			bit2	Bi-fuel Switch in Progress
			bit1	LV_GAS_SYSTEM_SAFE
			bit0	LV_INJECTION_ACTIVE

C.2.196 DID 0x0425-Status_8

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_8	BIN	bit7	unused
			bit6	unused
			bit5	LV_FUEL_TYPE_4
			bit4	LV_FUEL_TYPE_3
			bit3	LV_FUEL_TYPE_2
			bit2	LV_FUEL_TYPE_1
			bit1	unused
			bit0	LV_FUEL_TYPE_ECU

C.2.197 DID 0x0426-CONF_ENG

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
CONF_ENG	1	-	1	0…FFH	0…255

C.2.198 DID 0x0427-ERR_GAS

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
ERR_GAS	1	-	1	0…FFH	0…255

C.2.199 DID 0x0428-T_PEAK_IV_GAS_TI

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
T_PEAK_IV_GAS_TI	2	Ms	0.004	0…FFFFH	0…262.14

C.2.200 DID 0x0429-TI_ADD_DLY

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TI_ADD_DLY	2	Ms	0.004	8000…7FFFH	-131.072…131.068

C.2.201 DID 0x042A-TGAS_L

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TGAS_L	1	° C	0.75	0…FFH	-48…143.25

C.2.202 DID 0x042B-TGAS_H

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TGAS_H	1	° C	0.75	0…FFH	-48…143.25

C.2.203 DID 0x042C-PGAS_L

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits .
PGAS_L	2	hPa	0.25	0…FFFFH	0…16383.75

C.2.204 DID 0x042D-PGAS_H

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
PGAS_H	2	hPa	6	0…FFFFH	0…393210

C.2.205 DID 0x042E-FTL_AFU

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
FTL_AFU	1	%	0.390625	0…FFH	0…99.609375

C.2.206 DID 0x042F-TRT

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
TRT	4	H	2.777e-005	0…FFFFFFFFH	0…119304.647083

C.2.207 DID 0x0430-TRT_AFU

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Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits Max.
TRT_AFU	4	H	2.777e-005	0…FFFFFFFFH	0…119304.647083

C.2.208 DID 0x0431-Status_9

Byte No.	Description	Data Type	Value (Hex)	
data #1	Status_9	BIN	bit7	Unused
			bit6	Unused
			bit5	Unused
			bit4	Unused
			bit3	LV_SOV_GAS_H_STATE_3
			bit2	LV_SOV_GAS_H_STATE_2
			bit1	LV_SOV_GAS_H_STATE_1
			bit0	LV_SOV_GAS_PCD_STATE

C.2.209 DID 0x0432-MASS_GAS_PERC

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
MASS_GAS_PERC	2	Kg	0.001777523	0…FFFFH	0…116.49

C.2.210 DID 0x0500-Cruise Status

Byte No.	Description	Data Type	Value (Hex)
data #1	Cruise Status	Hex	0x00: Cruise OFF 0x01: Cruise ON

C.2.211 DID 0x0501- Cruise speed

Data Name	#of bytes	Unit	Scaling	HEXlimits	Physicallimits
Cruise speed	2	Km/h	0.01	0…FFFFH	0…655.35

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Annex D (normative) Stored data transmission functional unit data parameter definitions

The detailed description about this Annex can be found in Annex D of the document *CN200S General Diagnostic Requirement Specification*.

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Annex E (normative) Input output control functional unit data parameter definitions

E.1 InputOutputControlParameter definitions

The detailed description about this Annex can be found in Annex E.1 of the document *CN200S General Diagnostic Requirement Specification*.

It is possible for ECM supplier to add the additional definitions in Annex E.1 for CN200S vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

E.2 InputOutputControlByIdentifier (2F hex) controlOptionRecord definitions

E.3 InputOutputControlByIdentifier (2F hex) controlEnableMaskRecord definitions

N/A

E.4 InputOutputControlByIdentifier (2F hex) controlStatusRecord definitions

The definition of controlStatusRecord is the same as controlOptionRecord which defined in chapter E.2 of Annex E.

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Annex F (normative) Remote activation of routine functional unit data parameter definitions

F.1 RoutineIdentifier definition

The detailed description about this Annex can be found in Annex F.1 of the document *CN200S General Diagnostic Requirement Specification*.

It is possible for ECM supplier to add the additional definitions in Annex F.1 for CN200S vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

F.2 RoutineControl (31 hex) routineControlOption definitions

The following table lists the definition about routine control DID, according to the requirements of the document *CN200S General Diagnostic Requirement Specification*, the range which ECM used is 0x0200~0xDFFF.

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Annex G (informative) Examples for addressAndLengthFormatIdentifier parameter values

The detailed description about this Annex can be found in Annex G of the document *CN200S General Diagnostic Requirement Specification*.

It is possible for ECM supplier to add the additional definitions in Annex G for CN200S vehicle, if necessary, it is ECM supplier's responsibility to add the definitions, and obtain the confirmation from SGMW.

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